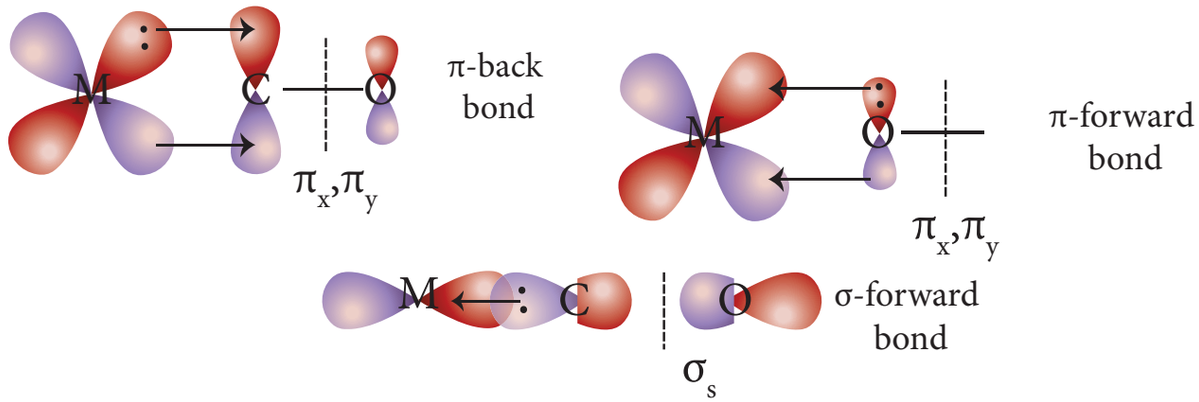
is called π-back bonding . Thus in metal carbonyls, electron density moves from ligand to metal through sigma bonding and from metal to ligand through pi bonding, this synergic effect accounts for strong $M \leftarrow CO$ bond in metal carbonyls. This phenomenon is shown diagrammatically as follows.



5.7 Stability of metal complexes:

The stability of coordination complexes can be interpreted in two different ways. The first one is thermodynamic stability and second one is kinetic stability. Thermodynamic stability of a coordination complex refers to the free energy change (ΔG) of a complex formation reaction. Kinetic stability of a coordination complex refers to the ligand substitution. In some cases, complexes can undergo rapid ligand substitution; such complexes are called labile complexes. However, some complexes undergo ligand substitution very slowly (or sometimes no substitution), such complexes are called inert complexes.

Stability constant:(β)

The stability of a coordination complex is a measure of its resistance to the replacement of one ligand by another. The stability of a complex refers to the degree of association between two species involved in an equilibrium. Let us consider the following complex formation reaction

$$Cu^{2+} + 4NH_3 \rightleftharpoons \left[Cu(NH_3)_4\right]^{2+}$$

$$\beta = \frac{\left[Cu(NH_3)_4\right]^{2+}}{\left[Cu^{2+}\right]\left[NH_3\right]^4} \qquad \text{---------(1)}$$

So, as the concentration of $\left[Cu(NH_3)_4\right]^{2+}$ increases the value of stability complexes also increases. Therefore the greater the value of stability constant greater is the stability of the complex.

Generally coordination complexes are stable in their solutions; however, the complex ion can undergo dissociation to a small extent. Extent of dissociation depends on the strength of the metal ligand bond, thus Stronger the $M \leftarrow L$, lesser is the dissociation.

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